

SIDDHARTH BHARDWAJ

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SUMMARY

Full-Stack Software Engineer with experience building scalable distributed systems, cloud-native applications, AI-powered products, and modern web applications using React, Next.js, FastAPI, Spring Boot, PostgreSQL, and AWS. Experienced across the entire software lifecycle from frontend UX design to backend architecture, infrastructure automation, and security engineering.

SKILLS

Systems & Languages: C/C++, Python, JavaScript, TypeScript, Java, Bash, SQL

Backend: FastAPI, Django, Express, Flask, Microservices, Spring Boot, Node.js, Apache Kafka, RabbitMQ, REST APIs

Databases: MySQL, MongoDB, Redis, PostgreSQL, Elasticsearch, Distributed Databases

Cloud & DevOps: Docker, Terraform, CloudFormation, Kubernetes, AWS, Google Cloud Platform (GCP), Git, Redis, CI/CD Pipelines

Frontend: React, Redux, React Native, Next.js, React Native, Tailwind CSS, TypeScript, Javascript, HTML/CSS

AI & Productivity: Cursor, OpenClaw, GitHub Copilot, LLM Integration (LangChain), Automated Documentation, Prompt Engineering

PROJECTS

Privacy Guardian – AI-Powered Browser Privacy Assistant

React, TypeScript, FastAPI, PostgreSQL, LLaMA 3B, Chrome Extensions

- Built a full-stack browser extension that helps users improve online privacy through real-time security recommendations, adaptive quizzes, and personalized learning. The platform combines a React-based dashboard, FastAPI backend services, and a fine-tuned LLaMA model to analyze browsing behavior, detect privacy risks, and generate contextual guidance.

MedHub – Healthcare Management Platform

React, Spring Boot/FastAPI, PostgreSQL, Docker

- Developed a full-stack healthcare management platform that streamlines patient, doctor, and appointment workflows through a centralized web application. Built responsive React interfaces and scalable backend APIs to support appointment scheduling, medical record management, user authentication, and real-time healthcare data access.

EXPERIENCE

HackerOne | *Freelancing Security Researcher*

Dec 2025 - Present

- **Responsible Disclosure:** Patched a critical XSS vulnerability for Reddit (50M+ DAU) by implementing a more robust, distributed session integrity check that neutralized session hijacking risks
- **Vulnerability Remediation:** Identified and engineered a fix for a critical integer overflow within QuestDB's core engine, preventing arithmetic instability and ensuring data consistency in high-throughput production clusters
- **System Integrity:** Disclosed a critical flaw in the PayPal Client API that allowed for unsanitized input, leading to a complete overhaul of the API's validation middleware to protect millions of concurrent users

Maxlinear Tech | *Software Engineer*

Nov 2022 - Sep 2023

- **Design System Orchestration:** Designed a distributed control plane and architected a modular **React component** library using **Tailwind CSS** for next-generation **SoC platforms**, utilizing **Java NDK** and C++ to handle real-time data ingestion from 1,000+ edge nodes with sub-millisecond jitter
- **DevOps Integration:** Spearheaded the containerization of the frontend build pipeline using **Docker**, ensuring sub-60-second hot-reloads and environment parity for a cross-functional team of 20+
- **Performance Tuning:** Spearheaded deep-dive memory management audits and kernel-level debugging, reducing system-wide latency spikes by **40%** and ensuring **99.9%** uptime during mass-production stress tests measured using **IxChariot**
- **Workflow Optimization:** Integrated GitHub Copilot and custom AI prompts into the internal dev-tooling suite, accelerating the generation of unit tests and boilerplate code by **30%**.

AIDatabases | *Distributed Systems Engineer*

Jun 2022 - Oct 2022

- **Microservices Orchestration:** Architected a high-availability microservices ecosystem using **Java (Spring Boot)** and **Docker**, implementing a circuit-breaker pattern that slashed **P95 latency** from **3.2s** to **1.3s**.
- **Scalable Data Pipelines:** Engineered event-driven ETL pipelines using Apache Kafka and Spark, processing over 5TB of raw data daily across distributed clusters with zero data loss.
- **Database Scaling:** Led the transition from **monolithic storage** to a **sharded PostgreSQL** environment, implementing advanced indexing and read-replicas to support a **500% increase in concurrent query traffic**.

Hewlett Packard Enterprise | *Backend Engineering Intern*

Jan 2022 - May 2022

- **Infrastructure as Code (IaC):** Developed automated deployment tools using **Python** and **Terraform** to orchestrate **microservice** environments on **AWS**, reducing manual provisioning time by 85%.
- **CI/CD Optimization:** Integrated automated **unit and integration testing** into the deployment pipeline, ensuring that 95% of critical system regressions were caught before reaching the staging environment.

EDUCATION

Master of Science (MS), Computer Science | *Dalhousie University, Canada*

Canada Sep 2023 - Jun 2025

Bachelor of Technology (BTech), Computer Engineering | *Amity University, India*

India Jul 2018 - May 2022